

# swtpcsim — FLEX9 on the SWTPC 6809

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The built-in `swtpc09` machine is an SWTPC 6809: the MP-09 processor board with the S-BUG monitor, 56K of RAM, an MP-S console and a DC-4 floppy controller ( `dc4` ). Each machine file here is that machine with a TSC **FLEX9 2.8:3** boot disk mounted on drive 0, so it comes up at S-BUG's `>` prompt ready to boot:

```
swtpcsim examples/flex9/flex9-40.toml
S-BUG 1.8 - 56K
>U                               (S-BUG mini-floppy boot)
```

`U` is S-BUG's mini-floppy boot command. It restores drive 0, reads track 0 sector 1 into `$C000` and jumps there. FLEX signs on, asks the date, and drops to its `+++` prompt:

```
Timer not available.

FLEX - Version 2.8:3 - 56K

Date (MM,DD,YY)? 1,15,80

+++CAT
```

`Timer not available.` is FLEX's own message: the machine has no timer board, and FLEX runs without one. From `+++`, `CAT` lists the disk and the usual FLEX commands work.

## The three disks

The same FLEX9 in the three geometries the DC-4 supports. The board finds the geometry from the image's **size**, so nothing in a `.toml` sets tracks or sides:

| Machine file               | Disk                           | Geometry                    |
|----------------------------|--------------------------------|-----------------------------|
| <code>flex9-35.toml</code> | <code>FLEX9-2.83-35.DSK</code> | 35-track single-sided, 88K  |
| <code>flex9-40.toml</code> | <code>FLEX9-2.83-40.DSK</code> | 40-track single-sided, 100K |
| <code>flex9-DS.toml</code> | <code>FLEX9-2.83-DS.DSK</code> | 40-track double-sided, 200K |

All three come from Mike Douglas's *PC2Flop and Flop2PC (for SWTPC DC-x Controllers in SWTPC 6809)* on [deramp.com](http://deramp.com):

<https://deramp.com/downloads/swtpc/software/FLEX/6809%20FLEX/Disk%20Image%20Transfer/>

## Read-only by default

Drive 0 is mounted `writprotect = true`, so looking around FLEX can never change the image. To save files back (FLEX's `SAVE`, `NEWDISK`, a file copy), remove the `writprotect` line from the drive in the `.toml`, or copy the `.DSK` somewhere writable first.